Lecture 21

Conics

Example: Transform the conic $y^2 + 6y + 2x + 1 = 0$ into standard form, then find center, vertex, focus, directrix, length of latus rectum, eccentricity of the conic and sketch it. Also find the equation of Latus rectum.

Solution: Given,

$y^2 + 6y + 2x + 1 = 0$

$\Rightarrow y^2 + 2 \cdot y \cdot 3 + 9 - 9 + 2x + 1 = 0$

$\Rightarrow y^2 + 2 \cdot y \cdot 3 + 3^2 - 8 + 2x = 0$

$\Rightarrow (y + 3)^2 = -2x + 8$

$\Rightarrow (y + 3)^2 = -2(x - 4)$

So, it is a parabola.

Comparing the equation with $(y - k)^2 = -4p(x - h)$, we get,

$-4p = -2$

$\Rightarrow p = \frac{-2}{-4} = \frac{1}{2}$

$\therefore$ Center, $(h, k) = (4, -3)$.

For parabola, vertex = center.

$\therefore$ Vertex $(4, -3)$.

Focus: $(h - p, k)$

$$= \left(4 - \frac{1}{2}, -3\right)$$

$$= \left(\frac{7}{2}, -3\right)$$

Directrix: $x - h = p$

$$\Rightarrow x = p + h = \frac{1}{2} + 4$$

$$\Rightarrow x = \frac{1 + 8}{2}$$

$$\therefore x = \frac{9}{2}$$

Length of latus rectum: $|4p| = \left|4 \cdot \frac{1}{2}\right|$

$$= |2|$$

$$= 2$$

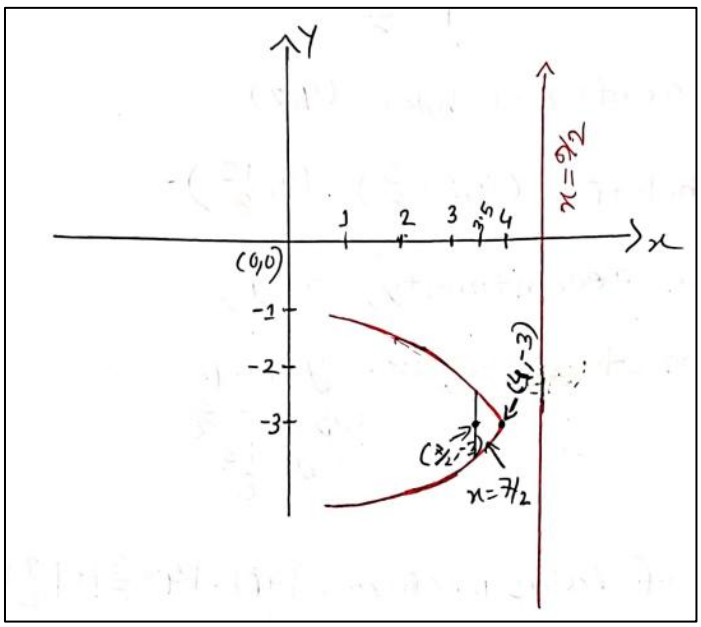
For parabola, eccentricity, $e = 1$

Equation of Latus Rectum, $x - h = -p$

$$\Rightarrow x = -p + h = -\frac{1}{2} + 4$$

$$\Rightarrow x = \frac{-1+8}{2}$$

$$\therefore x = \frac{7}{2}$$



H.W. Transform the conic $2x^2 - 16x - 3y + 38 = 0$ into standard form, then find center, vertex, focus, directrix, length of latus rectum, eccentricity of the conic and sketch it. Also find the equation of Latus rectum.

Answer: Standard Form: $(x-4)^2 = \frac{3}{2}(y-2)$

Vertex = Center, $(4,2)$

Focus, $= \left(4, \frac{19}{8}\right)$

Eccentricity, $e = 1$

Directrix, $y = \frac{13}{8}$

Length of Latus Rectum $= \frac{3}{2}$

The Equation of Latus Lectum, $y = \frac{19}{8}$

Example: Transform the conic $4x^2 + 25y^2 - 50y = 75$ into standard form, then find vertex, center, directrix, foci, eccentricity, length of latus rectum, major axis and minor axis. Then sketch it. Also find the equation of latus rectum.

Solution: Given,

For ellipse, we assume that the greater coefficient is 'a'.

$4x^2 + 25y^2 - 50y = 75$

$\Rightarrow 4x^2 + 25(y^2 - 2y) = 75$

$\Rightarrow 4x^2 + 25(y^2 - 2 \cdot y \cdot 1 + 1^2) - 25 = 75$

$\Rightarrow 4x^2 + 25(y - 1)^2 = 75 + 25$

$\Rightarrow 4x^2 + 25(y - 1)^2 = 100$

$\Rightarrow \frac{4x^2}{100} + \frac{25(y-1)^2}{100} = 1$

$\Rightarrow \frac{x^2}{25} + \frac{(y-1)^2}{4} = 1$

$\Rightarrow \frac{x^2}{5^2} + \frac{(y-1)^2}{2^2} = 1$

It is an equation of ellipse.

Comparing the equation with $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$, we get,

$h = 0, k = 1, a = 5, b = 2.$

$\therefore$ **Center,** $(h, k) = (0,1)$.

Vertices, $(h \pm a, k) = (\pm 5,1)$. So, the vertices are $(5,1)$ and $(-5,1)$.

We know,

$c^2 = a^2 - b^2$

$\Rightarrow c^2 = 5^2 - 2^2$

$\Rightarrow c^2 = 25 - 4$

$\Rightarrow c^2 = 21$

$\therefore c = \sqrt{21}$

$\therefore$ **Foci,** $(h \pm c, k) = \left(0 \pm \sqrt{21}, 1\right)$. So, the foci are $\left(\sqrt{21}, 1\right)$ and $\left(-\sqrt{21}, 1\right)$.

Equation of Directrix,

$$x - h = \pm \frac{a^2}{c}$$

$$\Rightarrow x = h \pm \frac{a^2}{c}$$

$$\Rightarrow x = 0 \pm \frac{5^2}{\sqrt{21}} = \pm \frac{25}{\sqrt{21}}$$

$$\therefore x = \pm \frac{25}{\sqrt{21}}$$

Eccentricity, $e = \frac{\sqrt{a^2-b^2}}{a} = \frac{\sqrt{5^2-2^2}}{5}$

$$= \frac{\sqrt{25-4}}{5}$$

$$= \frac{\sqrt{21}}{5}$$

Length of Latus Rectum $= \frac{2b^2}{a} = \frac{2\times2^2}{5} = \frac{2\times4}{5} = \frac{8}{5}$

Length of Major axis $= 2a = 2 \times 5 = 10$

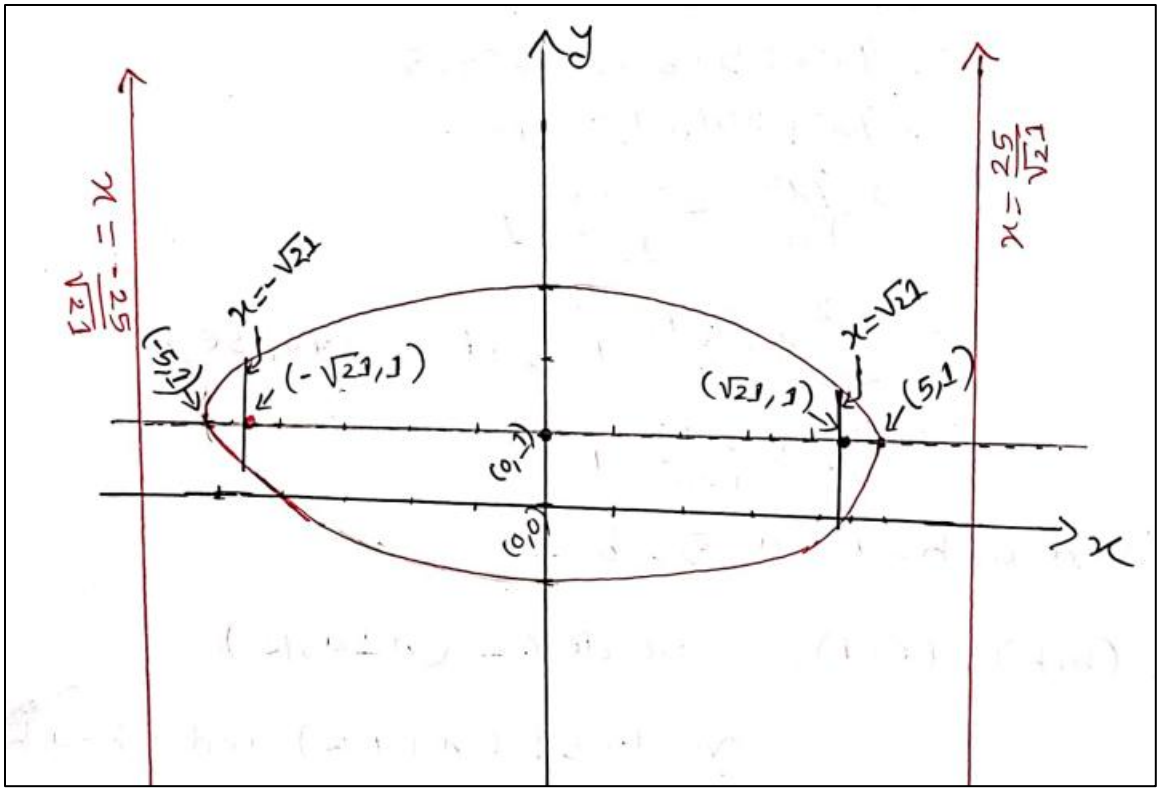
Length of Minor axis $= 2b = 2 \times 2 = 4$

Equation of Latus Rectum, $x - h = \pm c$

$$\Rightarrow x = h \pm c$$

$$\Rightarrow x = 0 \pm \sqrt{21}$$

$$\therefore x = \pm\sqrt{21}$$



H.W. Transform the conic $9x^2 - 54x + y^2 + 2y + 46 = 0$ into standard form, and then find center, vertices, foci, directrix, eccentricity, length of latus rectum, equation of the latus rectum and sketch the graph.

Answer: Standard Form: $\frac{(x-3)^2}{2^2} + \frac{(y+1)^2}{6^2} = 1$

Center, $= (3, -1)$

Vertices, $= (3,5)$ and $(3, -7)$

Foci, $= \left(3, -1 + 4\sqrt{2}\right)$ and $\left(3, -1 - 4\sqrt{2}\right)$

Eccentricity, $e = \frac{2\sqrt{2}}{3}$

Directrix, $y = -1 + \frac{36}{4\sqrt{2}}$ and $y = -1 - \frac{36}{4\sqrt{2}}$

Length of Latus Rectum $= \frac{4}{3}$, **major axis** = 12, **minor axis** = 4.

The Equation of Latus Lectum, $y = -1 + 4\sqrt{2}$ and $y = -1 - 4\sqrt{2}$

Example: Transform the conic $9y^2 - 4x^2 - 36y - 8x = 4$ into standard form, and then find center, vertices, foci, directrix, eccentricity, length of latus rectum, asymptotes of the conic and sketch the graph.

Solution: Given,

$9y^2 - 4x^2 - 36y - 8x = 4$

$\Rightarrow 9(y^2 - 4y) - 4(x^2 + 2x) = 4$

$\Rightarrow 9(y^2 - 2 \cdot y \cdot 2 + 2^2) - 36 - 4(x^2 + 2 \cdot x \cdot 1 + 1^2) + 4 = 4$

$\Rightarrow 9(y - 2)^2 - 4(x + 1)^2 = 4 - 4 + 36$

$\Rightarrow \dfrac{9(y-2)^2}{36} - \dfrac{4(x+1)^2}{36} = 1$

$\Rightarrow \dfrac{(y-2)^2}{4} - \dfrac{(x+1)^2}{9} = 1$

$\therefore \dfrac{(y-2)^2}{2^2} - \dfrac{(x+1)^2}{3^2} = 1$

For hyperbola, we assume that the positive coefficient is 'a'.

So, it is an equation of a hyperbola.

Comparing the equation with $\frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1$, we get,

$k = 2, h = -1, a = 2, b = 3.$

$\therefore$ **Center,** $(h, k) = (-1,2)$.

$\therefore$ **Vertices,** $(h, k \pm a) = (-1, 2 \pm 2)$

$\Rightarrow (h, k \pm a) = (-1, 2 + 2)$ and $(-1, 2 - 2)$

$\Rightarrow (h, k \pm a) = (-1, 4)$ and $(-1, 0)$

So, the vertices are $(-1,4)$ and $(-1,0)$.

We know that,

$c^2 = a^2 + b^2$

$\Rightarrow c^2 = 2^2 + 3^2 = 4 + 9 = 13$

$\Rightarrow c^2 = 4 + 9$

$\Rightarrow c^2 = 13$

$\therefore c = \sqrt{13}$

$\therefore$ **Foci,** $(h, k \pm c) = \left(-1, 2 \pm \sqrt{13}\right)$

$\Rightarrow (h, k \pm c) = \left(-1, 2 + \sqrt{13}\right)$ and $\left(-1, 2 - \sqrt{13}\right)$

Thus, the foci are $\left(-1,2+\sqrt{13}\right)$ and $\left(-1,2-\sqrt{13}\right)$.

$\therefore$ **Directrix,** $y-k=\pm\frac{a^2}{c}$

$$\Rightarrow y=k\pm\frac{a^2}{c}=2\pm\frac{2^2}{\sqrt{13}}$$

$$\Rightarrow y=2\pm\frac{4}{\sqrt{13}}$$

Thus, the directrix are, $y=2+\frac{4}{\sqrt{13}}$ and $y=2-\frac{4}{\sqrt{13}}$.

Eccentricity, $e=\frac{\sqrt{a^2+b^2}}{a}$

$$=\frac{\sqrt{2^2+3^2}}{2}$$

$$=\frac{\sqrt{4+9}}{2}$$

$$=\frac{\sqrt{13}}{2}$$

Length of Latus Rectum $=\frac{2b^2}{a}=\frac{2\times3^2}{2}=9$

The Equation of Asymptote,

$$x-h=\pm\frac{b}{a}(y-k)$$

$$\Rightarrow (x+1)=\pm\frac{3}{2}(y-2)$$

Now,

$$x+1=\frac{3}{2}(y-2)$$

$$\Rightarrow\ 2x+2=3y-6$$

$$\Rightarrow 3y=2x+2+6$$

$$\Rightarrow y=\frac{2}{3}x+\frac{8}{3}$$

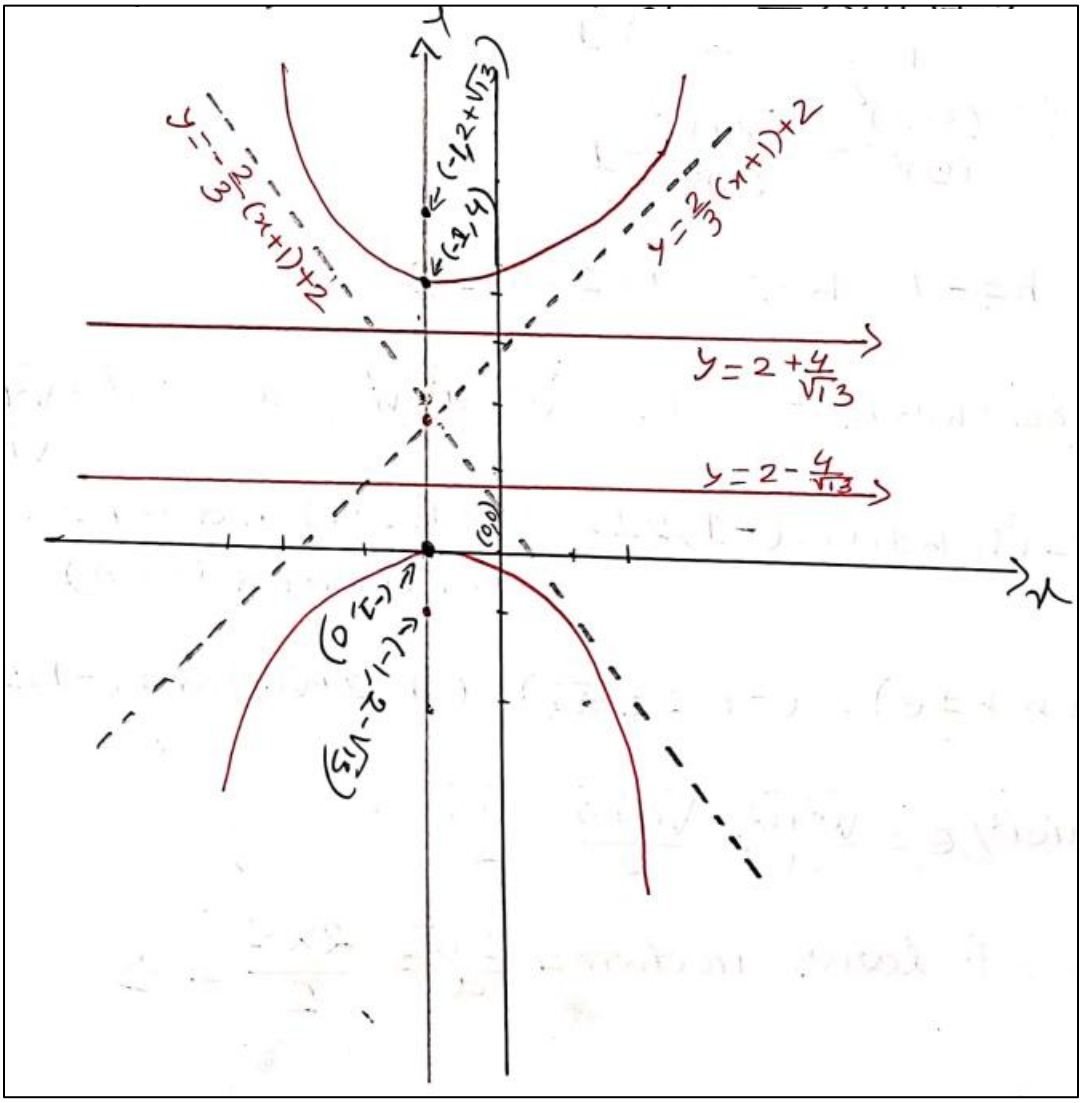
Also,

$$x+1=-\frac{3}{2}(y-2)$$

$$\Rightarrow\ 2x+2=-3y+6$$

$\Rightarrow 3y = -2x - 2 + 6$

$\Rightarrow y = -\frac{2}{3}x + \frac{4}{3}$



H.W. Transform the conic $x^2 - y^2 + 2y = 2$ into standard form, and then find center, vertices, foci, directrix, eccentricity, length of latus rectum, asymptotes of the conic, equation of the latus rectum and sketch the graph.

Answer: **Standard Form:** $\frac{x^2}{1^2} - \frac{(y-1)^2}{1^2} = 1$

Center, $= (0,1)$

Vertices, $= (1,1)$ and $(-1,1)$

Foci, $= (\sqrt{2}, 1)$ and $(-\sqrt{2}, 1)$

Eccentricity, $e = \sqrt{2}$

Directrix, $x = \frac{1}{\sqrt{2}}$ and $x = -\frac{1}{\sqrt{2}}$

Length of Latus Rectum $= 2$

The Equation of Asymptote, $y = x + 1$ and $y = -x + 1$

The Equation of Latus Lectum, $x = \sqrt{2}$ and $x = -\sqrt{2}$

H.W. Exercise 10.4 (3-14)